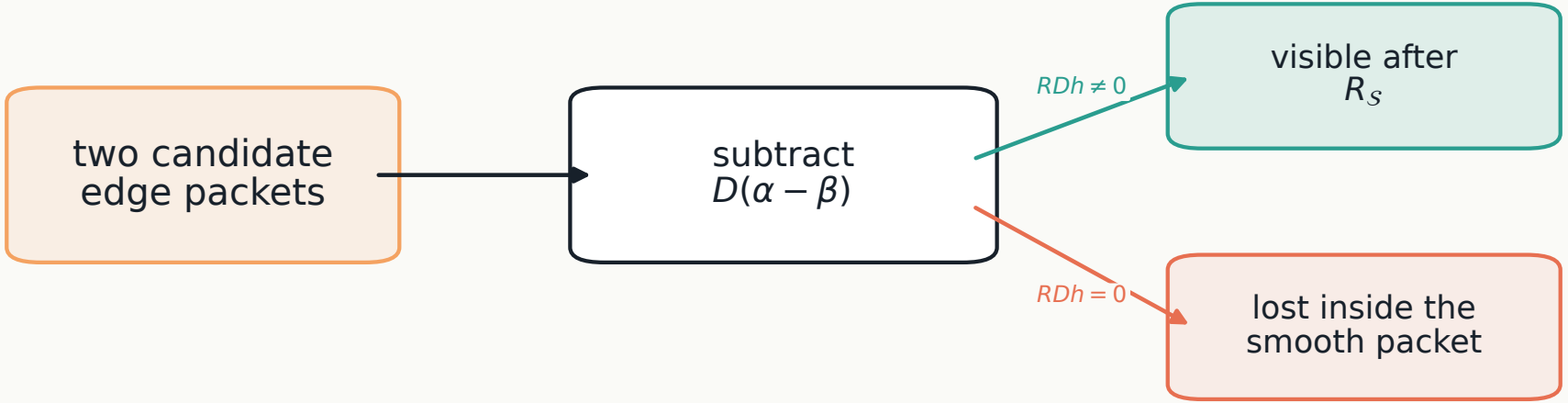

Universal s-sparse recovery holds exactly when no nonzero $2s$-sparse secant is hidden by the background.